

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0713

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

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Annexure A

Debugging & Optimisation Task

Code of Conduct:

*I **C.BHUVANESH (192521108)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:C.BHUVANESH

Annexure A

Debugging & Optimisation Task

- A rapidly growing IT company has shifted its operations to a newly constructed six-storey smart office building. More than **1,200 employees** use wireless laptops, smartphones, IP phones, printers, and IoT devices connected through the company's **Wireless Local Area Network (WLAN)**. The office relies heavily on cloud computing, online meetings, video conferencing, and file sharing, making uninterrupted wireless connectivity essential for daily operations. In addition, the company has a team of field engineers and sales executives who travel across different cities and use **4G/5G Wireless Wide Area Network (WWAN)** services to access the company's servers, customer databases, and business applications from remote locations.

During the last few weeks, employees have reported several connectivity issues. Users experience slow internet speeds during peak working hours, frequent Wi-Fi disconnections while moving between floors, and weak wireless signals in conference rooms. Video conferencing applications often freeze or disconnect, affecting communication with international clients. Network administrators have also observed that some wireless access points are overloaded while others remain underutilized, leading to uneven traffic distribution. Furthermore, field engineers working in rural and highway areas frequently lose WWAN connectivity, delaying the upload of maintenance reports and customer information. Security audits reveal that a few wireless devices are still connected using outdated security protocols, exposing sensitive company information to potential cyber threats. The management has requested the network engineering team to investigate these issues, identify the root causes, and optimize both the WLAN and WWAN infrastructure to ensure high performance, secure communication, and seamless connectivity for all users.

1. WLAN & WWAN Debugging and Optimisation

Problems Identified

- Slow internet during peak hours.
- Wi-Fi disconnects while moving between floors.
- Weak signal in conference rooms.
- Video conferencing freezes.
- Some Access Points (APs) are overloaded.
- Poor WWAN coverage in rural/highway areas.
- Outdated wireless security protocols.

Root Causes

- High user density causing AP congestion.
- Poor AP placement and overlapping channels.

- Weak signal strength and interference.
- Improper roaming configuration.
- Limited WWAN signal coverage.
- Use of outdated security (WEP/WPA).

Debugging Steps

- Perform wireless site survey.
- Check AP utilization.
- Analyze Wi-Fi channels for interference.
- Test roaming between APs.
- Measure WWAN signal strength.
- Scan network for insecure devices.

Optimisation Techniques

- Add more Access Points in crowded areas.
- Configure load balancing between APs.
- Use non-overlapping Wi-Fi channels (1, 6, 11 for 2.4 GHz).
- Enable fast roaming (802.11r/k/v).
- Upgrade to Wi-Fi 6/6E.
- Use dual-band (2.4 GHz & 5 GHz).
- Enable WPA3 security.
- Use VPN for WWAN users.
- Deploy signal boosters or multiple mobile carriers for remote users.

Expected Result

- Faster internet speed.
- Smooth roaming.
- Better signal coverage.
- Stable video conferencing.
- Improved security.
- Reliable remote connectivity.

2. A multinational financial organization operates its headquarters, regional offices, and multiple data centers using an **Asynchronous Transfer Mode (ATM) network** to support mission-critical services such as online banking, video conferencing, Voice over IP (VoIP), and real-time financial transactions. The ATM network uses fixed-size **53-byte cells** and establishes **Virtual Circuits (VCs)** to provide reliable and predictable Quality of Service (QoS). Recently, the organization has experienced several performance issues during peak business hours. Customers report delays in online transactions, voice calls suffer from jitter and occasional packet loss, and video conferencing sessions experience reduced quality. Network monitoring indicates that several ATM switches are operating at high utilization, some Virtual Circuits are congested, and available bandwidth is not

being efficiently allocated among different traffic classes. Engineers also discover that certain applications are using inappropriate ATM service categories, leading to increased delay and inefficient resource utilization. The IT department has assigned you as the network engineer to investigate the ATM network, identify the faults, debug the performance issues, and recommend optimization strategies to restore efficient, reliable, and high-quality communication across the enterprise.

Problems Identified

- Delay in online banking.
- Voice call jitter.
- Packet loss.
- Poor video quality.
- Congested Virtual Circuits.
- High ATM switch utilization.
- Incorrect ATM service categories.

Root Causes

- Congested Virtual Circuits (VCs).
- Inefficient bandwidth allocation.
- Incorrect QoS configuration.
- Wrong ATM service category selection.

Debugging Steps

- Monitor ATM switch utilization.
- Check VC congestion.
- Verify QoS settings.
- Measure delay, jitter, and packet loss.
- Analyze traffic flow.

Optimisation Techniques

- Redistribute traffic across multiple VCs.
- Allocate bandwidth according to traffic priority.
- Use proper ATM service categories:
 - CBR for VoIP
 - VBR for video
 - ABR for normal data
 - UBR for background traffic
- Upgrade overloaded ATM switches.

- Enable traffic shaping and congestion control.

Expected Result

- Lower delay.
- Reduced jitter.
- Better video quality.
- Improved bandwidth utilization.
- Stable financial transactions.

3. A multinational e-commerce company has interconnected its headquarters, regional offices, warehouses, and cloud data centers using a **Multiprotocol Label Switching (MPLS)** network to ensure fast, secure, and reliable communication. The MPLS infrastructure supports critical business applications, including online order processing, enterprise resource planning (ERP), video conferencing, Voice over IP (VoIP), cloud-based inventory management, and real-time customer support. Recently, the organization has experienced frequent performance issues during peak business hours. Employees report slow access to cloud applications, delayed order processing, poor-quality voice calls, and interruptions during video conferences. Network monitoring reveals that some Label Switched Paths (LSPs) are heavily congested, while other paths remain underutilized. The IT team also discovers that several routers have incorrect label mappings, causing inefficient routing and increased latency. In addition, traffic engineering has not been properly configured, resulting in uneven bandwidth utilization across the MPLS backbone. Since these issues are affecting customer satisfaction and business operations, the network administrator has asked you to investigate the MPLS network, identify the faults, debug the performance problems, and recommend suitable optimization techniques to improve the overall efficiency and reliability of the network.

Problems Identified

- Slow cloud applications.
- Delayed order processing.
- Poor VoIP quality.
- Video interruptions.
- Congested LSPs.
- Incorrect label mappings.
- Uneven bandwidth utilization.

Root Causes

- Incorrect MPLS labels.
- Poor traffic engineering.
- Uneven traffic distribution.

- Congested Label Switched Paths.

Debugging Steps

- Verify MPLS label tables.
- Check LSP utilization.
- Test routing paths.
- Monitor latency and bandwidth.

Optimisation Techniques

- Correct MPLS label mappings.
- Configure MPLS Traffic Engineering (TE).
- Balance traffic across multiple LSPs.
- Prioritize ERP, VoIP, and video traffic using QoS.
- Upgrade heavily loaded routers.
- Enable fast reroute.

Expected Result

- Faster cloud access.
- Improved voice quality.
- Better video conferencing.
- Balanced network traffic.
- Reduced latency.

4. A multinational logistics company operates an automated warehouse where barcode scanners, RFID readers, autonomous robots, and inventory management systems continuously exchange data over a computer network. Every day, thousands of shipment records, customer details, and inventory updates are transmitted between warehouse terminals and the central server. Recently, the company has observed that some shipment records are arriving with incorrect information, barcode scans occasionally fail, and inventory quantities in the database do not always match the physical stock. During peak hours, network monitoring reports an increase in transmission errors caused by electrical interference, signal attenuation, and network congestion. Although some corrupted frames are detected and retransmitted, others remain undetected, resulting in incorrect database entries and delayed order processing. The IT department suspects that the current error detection and correction mechanisms are not configured efficiently. As the network engineer, you are assigned to investigate the transmission errors, identify the weaknesses in the existing error control techniques, and recommend suitable debugging and optimization strategies to improve the reliability and accuracy of data communication.

Problems Identified

- Incorrect shipment records.
- Barcode scan failures.
- Inventory mismatch.
- Transmission errors.
- Corrupted frames.

Root Causes

- Electrical interference.
- Signal attenuation.
- Network congestion.
- Weak error detection methods.

Debugging Steps

- Monitor frame error rate.
- Identify noisy communication links.
- Analyze retransmission logs.
- Verify CRC and checksum results.

Optimisation Techniques

- Use CRC (Cyclic Redundancy Check).
- Implement Hamming Code for error correction.
- Use Automatic Repeat reQuest (ARQ).
- Improve network cabling.
- Reduce interference.
- Upgrade switches and network equipment.
- Apply congestion control.

Expected Result

- Accurate data transmission.
- Reduced transmission errors.
- Correct inventory records.
- Reliable barcode scanning.

5. A leading online education platform conducts live classes and online examinations for more than **15,000 students** across different locations. During examinations, the central server simultaneously sends question papers, images, videos, and answer submission acknowledgements to all students

over the network. The server is equipped with a high-speed **10 Gbps** connection, whereas many students access the system using slower broadband or mobile internet connections. During peak hours, students experience incomplete downloads, delayed page loading, duplicate packets, and repeated retransmissions. Some students receive the next set of data before their devices finish processing the previous one, causing receiver buffer overflow and application crashes. Network administrators also observe an excessive number of retransmissions, increased latency, reduced throughput, and poor bandwidth utilization. The current flow control mechanism is not effectively regulating the transmission rate between the sender and receivers. As the network engineer, you have been assigned to investigate the communication problems, identify the causes of inefficient flow control, debug the existing system, and recommend suitable optimization techniques to ensure reliable and efficient data transmission during high-traffic periods.

Problems Identified

- Incomplete downloads.
- Delayed page loading.
- Duplicate packets.
- Buffer overflow.
- Excessive retransmissions.
- High latency.
- Low throughput.

Root Causes

- Sender transmits faster than receiver.
- Receiver buffer overflow.
- Inefficient flow control.
- Heavy network congestion.

Debugging Steps

- Monitor sender and receiver speeds.
- Check buffer utilization.
- Analyze retransmission statistics.
- Measure network throughput and latency.

Optimisation Techniques

- Implement Sliding Window Protocol.
- Use TCP Flow Control.
- Apply congestion control algorithms.
- Increase receiver buffer size.

- Enable Selective Repeat ARQ.
- Use traffic shaping and load balancing.
- Prioritize examination traffic using QoS.

Expected Result

- No buffer overflow.
- Fewer retransmissions.
- Faster downloads.
- Higher throughput.
- Better bandwidth utilization.
- Smooth online examinations.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation